H

VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CLASS		TUTOR'S NAME	

PHYSICS 9478/02

Paper 2 Structured Questions **2 September 2026**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and tutor's name in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 9
2	/ 8
3	/ 9
4	/ 10
5	/ 10
6	/ 9
7	/ 20
Total	/ 75

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $(1 / (36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = F / A$
gravitational potential	$\phi = -Gm / r$
temperature	$T / \mathrm{K} = T / {}^\circ\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal molecule	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} CV^2$
charging a capacitor	$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

.....

.....

.....

.....[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

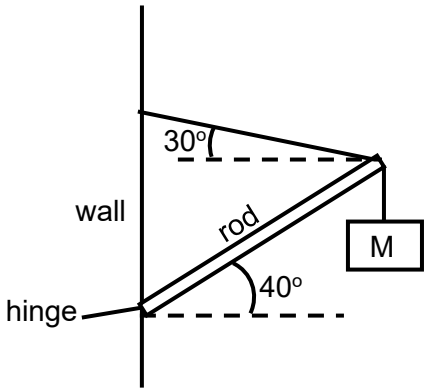


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]

2 A speedboat of mass 450 kg is equipped with two identical motors. Each motor can produce 75 kW of power. When both motors are turned on, its maximum speed is 30 m s^{-1}. Assume that the drag force on the speedboat is proportional to the square of its speed.

(a) Determine the maximum speed of the boat when only one motor is turned on.

maximum speed = ... m s^{-1} [3]

(b) When both motors are turned on, the speedboat takes 9.0 s to reach the maximum speed of 30 m s^{-1}. Determine the amount of energy that is lost to the surroundings during that time.

amount of energy lost = ... J [2]

(c) Sketch, on Fig. 2.1, the variation with time t of the speed of the speedboat v, until it reaches maximum speed.



Fig. 2.1

[1]

(d) When the speedboat is travelling at a speed of 20 m s^{-1}, both motors are turned off at the same time. The speedboat continues to travel an additional distance of 600 m before coming to rest. Determine the average resistive force acting on the speedboat.

average resistive force = ... N [2]

[Total: 8]

3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius *r*, as shown in Fig. 3.1.

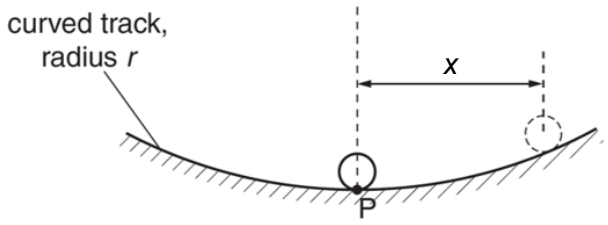


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

.....

.....

.....

.....[2]

(ii) The radius *r* of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

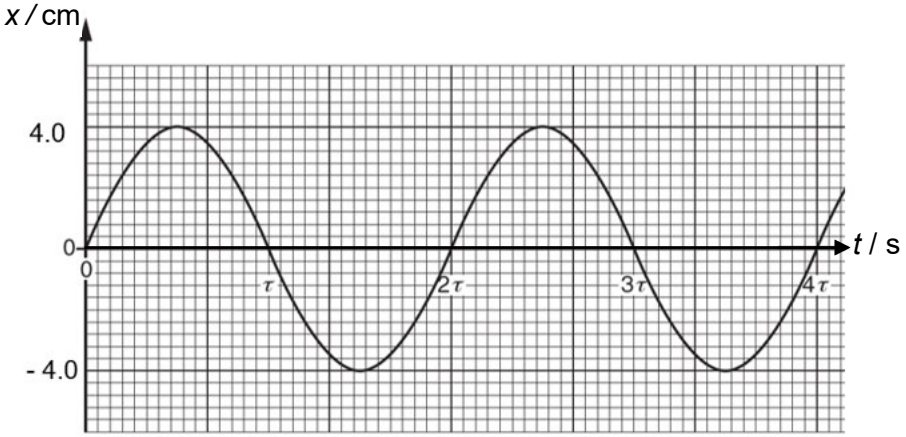


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

maximum kinetic energy= .....J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

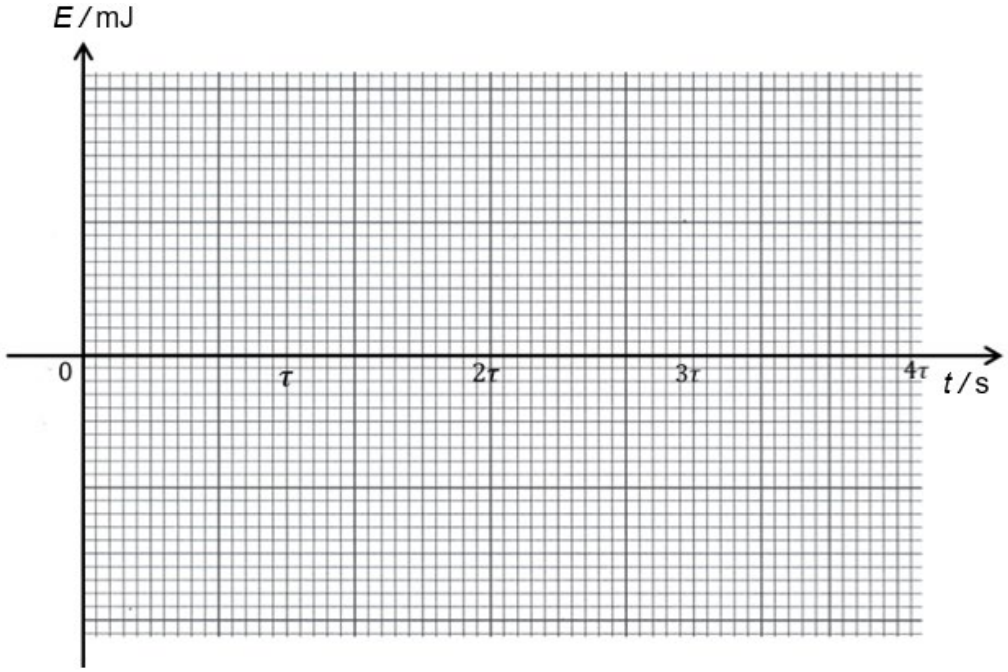


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

[Total: 9]

4 (a) (i) A cubic container of sides s and cross-sectional area A contains a gas made up of a large number of molecules, each with mass m. One of these molecules has an initial velocity c and a velocity component u_x in the x direction, as shown in Fig. 4.1.

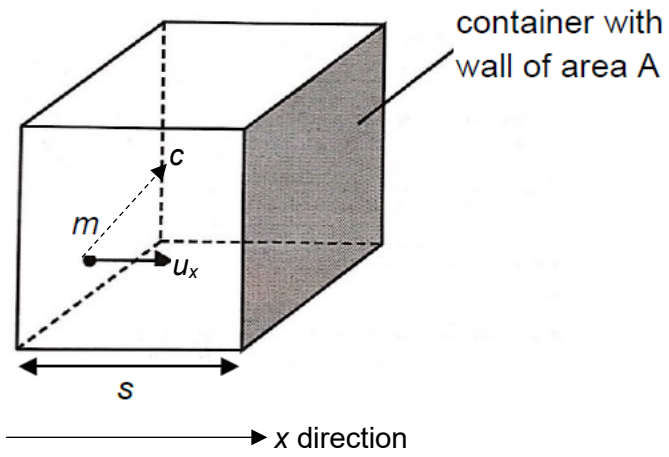


Fig. 4.1

Derive, in terms of m, u_x and s,

1. the change in momentum of the molecule when it collides with the wall on the right, [1]

2. the time between consecutive collisions with the wall on the right. [1]

(ii) Hence show that the average force F exerted by the molecule on the wall on the right is given by $F = \frac{mu_x^2}{s}$. [1]

(iii) Each molecule of the gas has its own velocity c and velocity component u_x in the x direction.

1. State the equation that relates the mean square speed of the molecules, $<c^2>$, to the mean square speed due their velocity components in the x direction, $<u_x^2>$. [1]

2. Using your answers to **(a)(ii) and (a)(iii)1**, derive the expression $pV = \frac{1}{3}Nm\langle c^2 \rangle$, where p is the pressure on the wall, V is the volume of the container, N is the number of molecules and $\langle c^2 \rangle$ is the mean square speed of the molecules. [2]

(b) A cylinder, shown in Fig. 4.2, fitted with a piston contains ideal gas at a constant temperature of 20 °C. When the volume V is 0.50×10^{-3} m^3, the pressure p in the cylinder is 20×10^4 Pa.

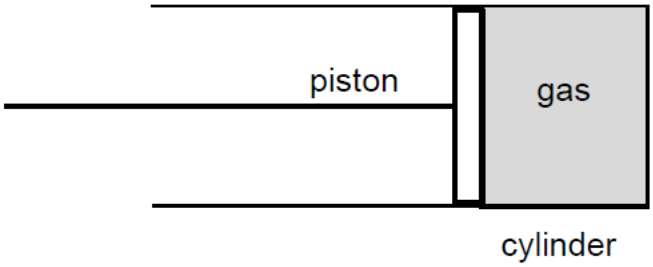


Fig. 4.2

An experiment was conducted and the data was plotted in Fig. 4.3

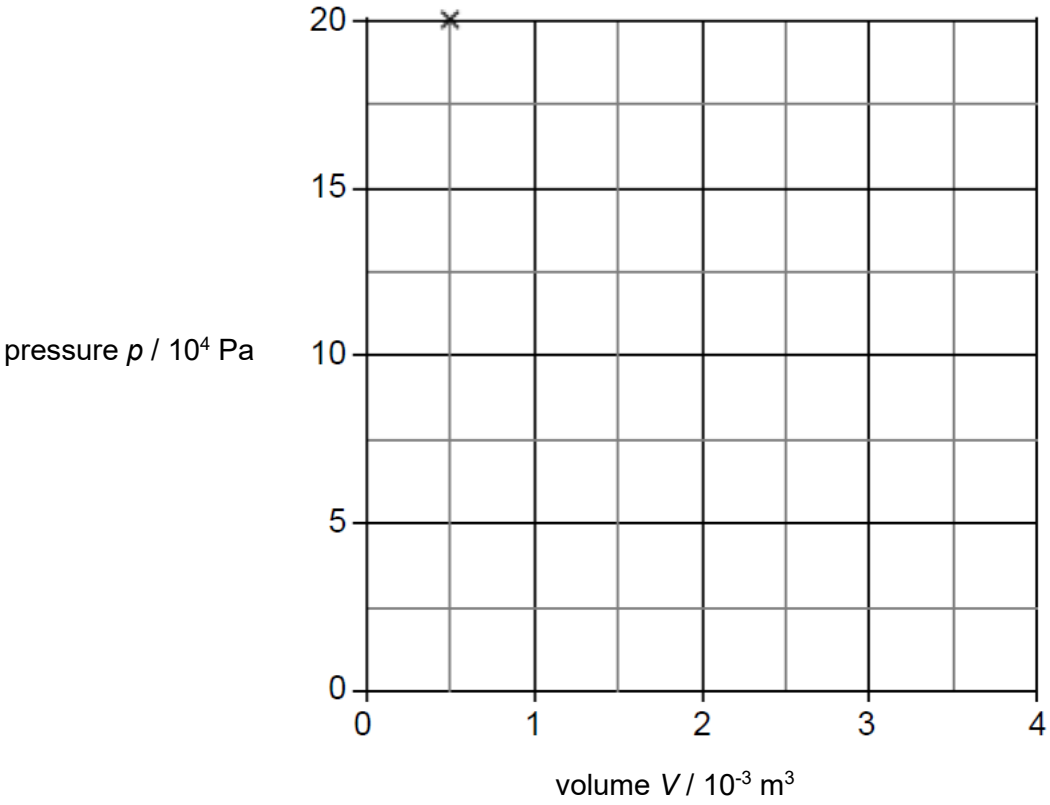


Fig. 4.3

(i) By plotting two additional points, draw a graph on Fig. 4.3 to show the relationship between p and V as the piston is slowly pulled out such that the temperature of the gas remains constant. [1]

(ii) Calculate the number of gas molecules in the cylinder.

number of gas molecules = ... [2]

(iii) Determine the total kinetic energy of the gas molecules.

total kinetic energy = ... J [1]

[Total: 10]

5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.

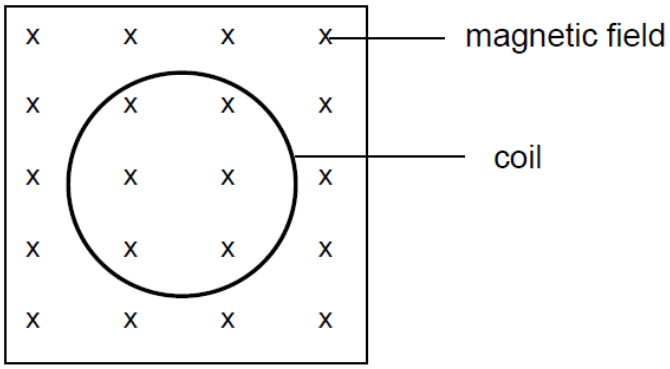


Fig. 5.1

(a) Define *magnetic flux*.

.....

.....

.....[1]

(b) **(i)** The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

.....

.....

.....[2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

.....

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.....[2]

(c) A magnet of length *x* and magnetic flux density *B* is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height *h* above the top of a *long* solenoid of length *L* and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet *x* is much smaller than the length of the solenoid *L*.

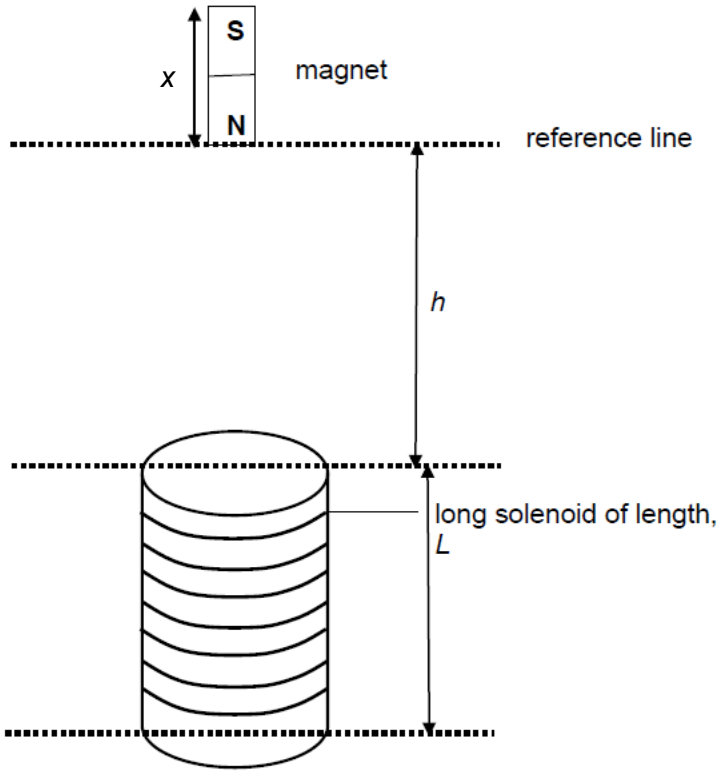


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)



Fig. 5.3

[2]

(ii) State how your graph in **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

.....

.....

.....[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

[2]

[Total: 10]

6 **(a)** A student claims that "Light is only a wave and not a particle."

Discuss whether this statement is correct, providing experimental evidence.

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.....[3]

(b) A cool, low-pressure gas of an element X is illuminated by a beam of monochromatic violet light of frequency 8.21×10^{14} Hz.

The simplified energy level diagram for an atom of element X is shown in Fig. 6.1 below.

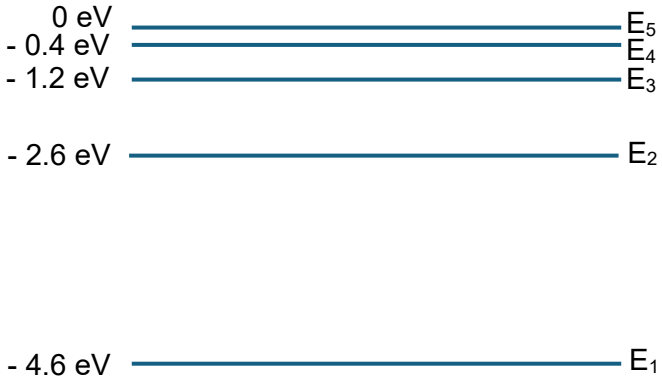


Fig. 6.1

(i) Calculate the energy of a photon of the incident violet light.

energy of a photon = ... eV [2]

(ii) Draw, on Fig. 6.1, the transition corresponding to the absorption of the incident violet photon. [1]

(iii) Determine the number of different emission lines that may be observed as a result of the absorption of many of such violet photons.

number of emission lines = ... [1]

(iv) Explain why the intensity of the violet light decreases after passing through the gas.

.....

.....

.....

.....

.....

.....[2]

[Total: 9]

7 Read the passage below and answer the questions that follow.

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator. It first started up on 10 September 2008 and is the crown jewel within the accelerator complex of the European Council for Nuclear Research (CERN).

Spanning across the Switzerland-France border, the LHC is a large horizontal ring in which high-energy particle beams, most commonly proton beams, are accelerated to speeds close to the speed of light in opposite directions. These proton beams circulate around the ring more than 11000 times a second as they are made to collide multiple times at different interaction points where the beams cross each other. By studying the fundamental particles produced in the aftermath of these proton-proton collisions, scientists hope to gain new insights and answers to some of the most important unsolved mysteries in Physics.

Inside the accelerator, the protons travel around the ring for hours along two separate beam pipes, both of which are kept at an ultrahigh vacuum. As the protons circulate around the LHC, they are repeatedly accelerated by strong electric fields along a short linear accelerator within the ring.

The protons are guided around the accelerator ring along the beam pipes by strong magnetic fields generated by thousands of electromagnets of different types and sizes, all of which need to be constantly kept at extremely low temperatures of about 1.9 K. This is why the LHC also houses the world's largest cryogenic system, using about 1.0×10^7 kg of liquid nitrogen and 1.3×10^5 kg of liquid helium to maintain the operation of the magnets.

The most common type of these electromagnets is called dipole magnets, which help bend the protons' trajectories so that they travel successfully along the curved beam pipes. These dipole magnets consist of many coils of electric cables placed next to and at many different sections along the beam pipe, each carrying a large electric current. The dipole magnets are contained and fastened together with strong non-magnetic stainless-steel collars.

Fig. 7.1 shows the cross-section of one of the dipole magnet setups at a section of a beam pipe of the LHC. The large number of electric cables are grouped into two double-layered regions around the beam pipe, labelled as region 1 and region 2 in Fig. 7.1. These electric cables are arranged parallel to the beam pipe such that when a current is travelling along these cables, the magnetic flux density produced by these cables at the centre of the beam pipe helps bend the protons' trajectories so that they can travel along the curved beam pipe.

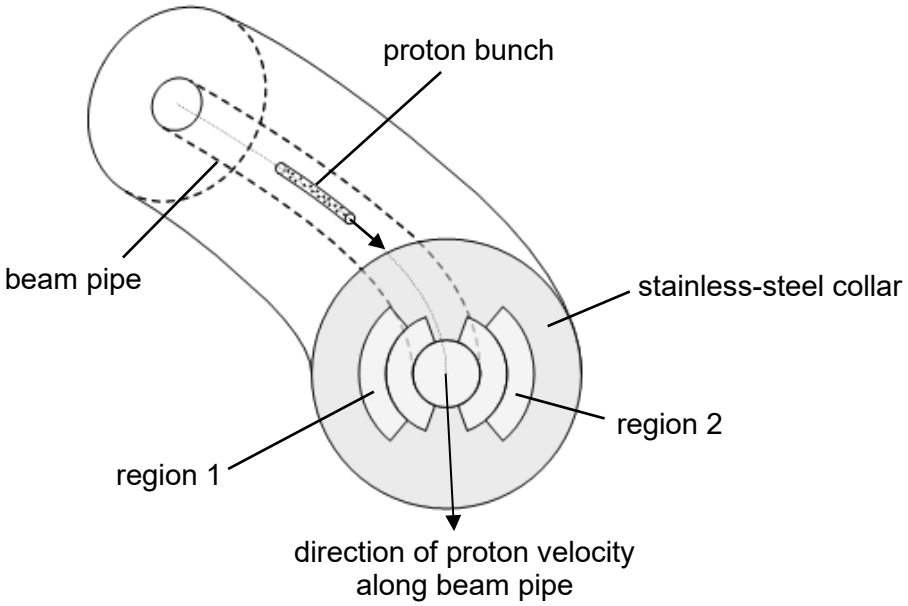


Fig. 7.1
(The electric cables parallel to the beam pipe are not shown for clarity.)

An important parameter in determining the functionality of a particle accelerator like the LHC is the beam current, which is defined as the average rate of flow of charge in the beam pipe. However, during normal operation, the protons do not travel in the beam pipe as a continuous beam. Instead, in each proton beam, the protons travel in groups called "bunches", which are approximately cylindrical in shape as shown in Fig. 7.1.

The LHC is built underground with a mean depth of 100 m for various reasons, including minimising the damage caused to the landscape and environment on the surface. In addition, protons travelling around the curved beam pipes accelerate to produce synchrotron radiation in the form of ultraviolet and X-ray photons, and interactions between the protons and the nuclei of any atoms around the beams in the beam pipes also produce ionising radiation. The Earth's crust therefore also provides natural shielding from these sources of radiation for anyone living on the surface near the LHC.

Table 7.1 shows some important data and parameters of the LHC under normal operating conditions.

Table 7.1

total circumference of LHC ring	26659 m
diameter of beam pipe	56 mm
proton energy	21.2 GeV
current in each cable of a dipole magnet setup	11850 A
length of a dipole magnet setup	14.3 m
peak dipole magnetic flux density at centre of beam pipe	8.33 T
beam current	0.58 A
number of bunches per proton beam	2808
length of one bunch	7.48 cm
cross-sectional area of one bunch	1.0 mm^2

(a) Suggest why the inside of the beam pipe needs to be maintained at an ultrahigh vacuum while the LHC is being operated.

.....

.....

.....[1]

(b) State and explain if an accelerator like the LHC can be used to study neutron-neutron collisions.

.....

.....

.....[1]

(c) (i) The protons in the LHC beam pipe travel at a speed of 0.999*c*, where *c* is the speed of light. Show that the period of revolution of the protons around the LHC ring is 8.89×10^{-5} s.

[1]

(ii) By considering the beam current, show that the total charge in the ring is 5.16×10^{-5} C.

[1]

(iii) Use your answer in **(c)(ii)** to determine the number of protons in each bunch within the proton beam.

number of protons = ... [2]

(d) Fig. 7.2 shows a close-up of the cross-section of the dipole magnet setup around the beam pipe in Fig. 7.1, where regions 1 and 2 are represented by combined cables 1 and 2 respectively. Each combined cable carries electric current of the same magnitude.

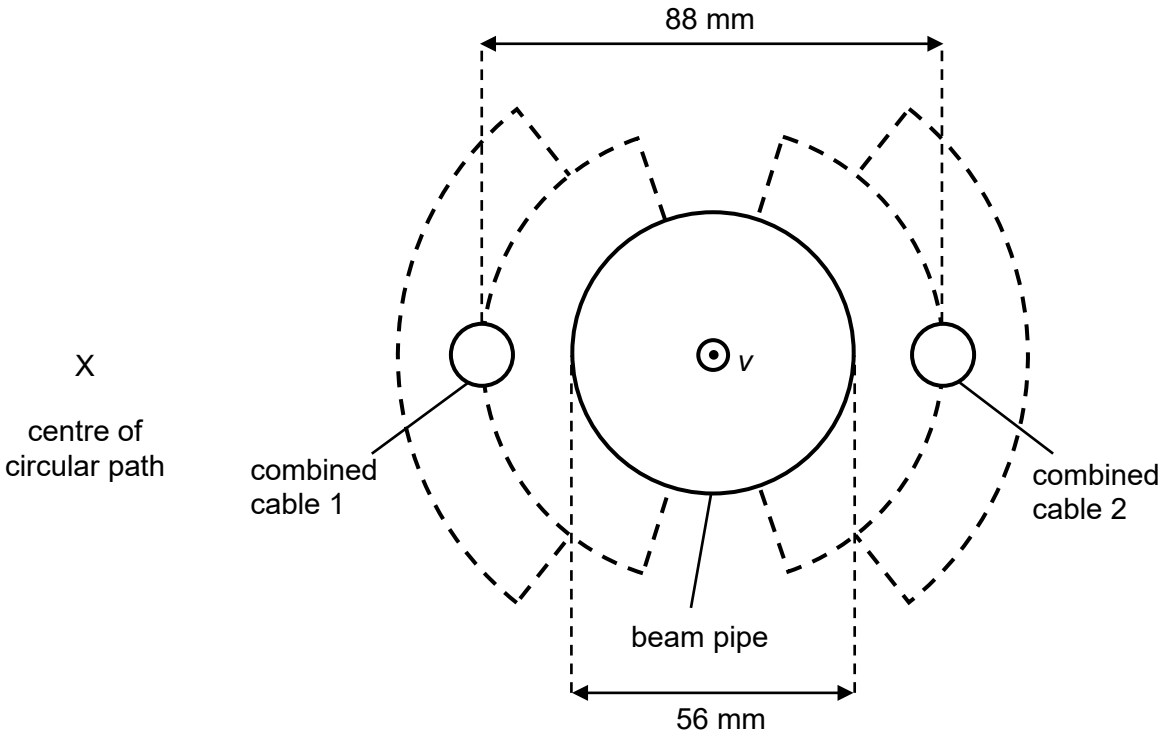


Fig. 7.2

In this model of the dipole magnet setup, the distance between the combined cables is 88 mm, while the diameter of the beam pipe is 56 mm.

The proton velocity *v* in the middle of the beam pipe is pointing out of the page, as shown in Fig. 7.2. The centre of the circular path of the proton's trajectory is on the side of combined cable 1.

(i) On Fig. 7.2, using appropriate symbols and labels, indicate the direction of

1. the magnetic flux density *B* at the centre of the beam pipe,

2. the currents I_1 and I_2 in the combined cables 1 and 2 respectively. [2]

(ii) 1. Determine the current required in each combined cable to produce the peak dipole magnetic flux density at the centre of the beam pipe.

current = ... A [2]

2. Hence, determine the number of cables in each region.

number of cables = ... [1]

3. Suggest a reason why the dipole magnets need to be contained and fastened together with collars that are strong.

Support your reason with appropriate calculations.

.....

.....

.....[3]

(e) Besides dipole magnets, another type of electromagnet used at the LHC are quadrupole magnets.

Quadrupole magnets help “squeeze” the travelling protons within each bunch closer together so that the protons stay travelling along the central axis of the beam pipe in a tightly focused beam. They do so by accounting for the electrostatic repulsion between the protons within each bunch.

(i) By approximating the average volume occupied by a proton in each bunch to be the volume of a cube, show that the average distance between two adjacent protons in a bunch is 8.66×10^{-7} m.

[2]

(ii) Hence determine the magnitude of the average repulsive force between two adjacent protons in a bunch.

average repulsive force = ... N [1]

(iii) The quadrupole magnets also help to keep the protons in the proton beam on their intended path by accounting for the protons falling downwards due to their weight.

Determine the number of rounds a proton will be able to travel around the LHC accelerator ring before falling to the bottom of the beam pipe due to gravity, if there were no quadrupole magnets.

Assume that the acceleration due to gravity in the pipe is 9.81 m s^{-2}.

number of rounds = ... [3]

[Total: 20]